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Operations Research

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Sample solution to the demo tasks 4

Demo Exercises

Exercise D4. 1. *Dual Problem*

Problem

Consider the following linear programs (LPs):

$$\begin{array}{lrrrl} \text{Max}\underset{\text{s.t.}}{} & 4x_1 - 7x_2 + 8x_3 & & & \\ & 3x_1 + + 2x_3 & \le & 18 \\ & 5x_1 + 7x_2 - x_3 & \ge & 5 \\ & -6x_1 + 4x_2 + 3x_3 & = & 30 \\ & x_1,\ x_2 & \ge & 0 \end{array}$$

$$\begin{array}{lrrrl} \text{Min}\underset{\text{s.t.}}{} & x_1 + 2x_2 - 6x_3 & & & \\ & 2x_1 + + 4x_3 & = & 12 \\ & - x_2 + 5x_3 & \ge & 10 \\ & -3x_1 + 3x_2 + x_3 & \le & 20 \\ & x_1,\ x_2 & \ge & 0 \end{array}$$

Determine the corresponding dual LPs.

Solution

We introduce a dual variable for each constraint, change the optimization operator, transpose the matrix A and swap the vectors b and c to obtain the form $Min/Max\ b^t y,\ s.t.\ A^t y\ ?\ c.$

After that, we still need to decide how the dual variables are defined and how to fill in the *?*. Here it is important to note:

- From equality constraints follow free variables (and vice versa).
- From normal constraints follow normal variables (and vice versa)
- From non-normal constraints follow non-normal variables (and vice versa)

$$\begin{array}{lrrr} \text{Min}\underset{\text{s.t.}}{} & 18y_1 + 5y_2 + 30y_3 & & \\ & 3y_1 + 5y_2 - 6y_3 & \ge & 4 \\ & 7y_2 + 4y_3 & \ge & -7 \\ & 2y_1 - y_2 + 3y_3 & = & 8 \\ & y_1 & \ge & 0 \\ & y_2 & \le & 0 \\ & y_3 & \in & \mathbb{R} \end{array}$$

$$\begin{array}{lrrr} \text{Max}\underset{\text{s.t.}}{} & 12y_1 + 10y_2 + 20y_3 & & \\ & 2y_1 + - 3y_3 & \le & 1 \\ & - y_2 + 3y_3 & \le & 2 \\ & 4y_1 + 5y_2 + y_3 & = & -6 \\ & y_1 & \in & \mathbb{R} \\ & y_2 & \ge & 0 \\ & y_3 & \le & 0 \end{array}$$

Exercise D4. 2. *Primal-Dual*

Problem

$$(P)\ \underset{\text{s.t.}}{\text{Max}}\ \ 2x_1 + 3x_2 + x_3 \qquad\qquad (D)\ \underset{\text{s.t.}}{\text{Min}}\ \ 4y_1 + 6y_2 + 8y_3$$

$$\begin{array}{rcl} 2x_1 + x_2 + x_3 & \leq & 4 \\ 4x_1 + 2x_3 & \leq & 6 \\ 5x_2 + 3x_3 & \leq & 8 \\ x & \in & \mathbb{R}^3_{\geq 0} \end{array} \qquad\qquad \begin{array}{rcl} 2y_1 + 4y_2 & \geq & 2 \\ y_1 + 5y_3 & \geq & 3 \\ y_1 + 2y_2 + 3y_3 & \geq & 1 \\ y & \in & \mathbb{R}^3_{\geq 0} \end{array}$$

a) Specify all primal and dual complementary slackness conditions for (P) and (D).

b) Let $x^* = (3/2, 1, 0)^\tau$ be a primal point. Use the complementary slackness conditions to check whether x^* is an optimal solution for (P).

c) Is the dual problem (D) unbounded, infeasible or does it have at least one finite optimal solution? Answer this question without solving the dual problem. Justify your answer.

Solution

a) Complementary slackness conditions:

$$\begin{array}{rr} (2x_1 + x_2 + x_3 - 4) \cdot y_1 = 0 & (2y_1 + 4y_2 - 2) \cdot x_1 = 0 \\ (4x_1 + 2x_3 - 6) \cdot y_2 = 0 & (y_1 + 5y_3 - 3) \cdot x_2 = 0 \\ (5x_2 + 3x_3 - 8) \cdot y_3 = 0 & (y_1 + 2y_2 + 3y_3 - 1) \cdot x_3 = 0 \end{array}$$

b) First, we check the feasibility of x^* by checking the constraints of (P) and obtain $[4 \leq 4], [6 \leq 6], [5 \leq 8]$. Thus, x^* fulfills all primal constraints and is a feasible point.

Since the 3rd primal constraint is not binding, we have $y_3 = 0$. Because $x_1, x_2 \neq 0$, $2y_1 + 4y_2 - 2 = y_1 + 5y_3 - 3 = 0$ must apply. By plugging in, we obtain $y_1 = 3$ and $y_2 = -1$.

However, since $y_2 \geq 0$ must hold, the point is not feasible. x^* is therefore not an optimal solution for (P).

c) Since the primal problem (P) has a feasible point with x^*, (D) cannot be unbounded (weak duality). Furthermore, it is easy to see that $y = (1, 1, 1)^\tau$ is a feasible point for (D). The dual problem is therefore not infeasible. It follows that (D) must have at least one finite optimal solution.

Exercise D4. 3. *Transportation Problem*

Problem

Consider the following transportation problem: You are responsible for distributing food supplies from collection points $i \in I$ to various food banks $j \in J$. Each collection point $i \in I$ has a supply capacity of food $s_i \geq 0$, and each food bank $j \in J$ has a demand $d_j \geq 0$ for food. Let $c_{ij} \geq 0$ denote the cost of transporting one unit of food from collection point $i \in I$ to food bank $j \in J$.

a) Formulate an LP (P) to determine a cost-minimizing transportation plan. Use variables $x_{ij} \geq 0$ for all $i \in I$, $j \in J$, where x_{ij} denotes the amount of food transported from collection point i to food bank j. Please ensure that (1) the demand at each food bank is fully met, and (2) no more food is transported from a collection point than is available.

b) Formulate the dual problem (D) corresponding to (P).

c) The dual problem can be interpreted as the optimization problem of an external transport company. What are the meanings of the dual variables in this context?

Solution

a) The LP (P) is given by:

$$
\begin{aligned}
(P) \quad \underset{\text{s.t.}}{\text{Min}} \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} & \\
\sum_{j \in J} x_{ij} \leq s_i, & \quad \forall i \in I \\
\sum_{i \in I} x_{ij} \geq d_j, & \quad \forall j \in J \\
x_{ij} \geq 0, & \quad \forall i \in I,\, j \in J
\end{aligned}
$$

b) Let us first multiply the inequalities for i by -1 so that the problem is in normal form. We introduce the dual variables π_i with $i \in I$ for the first part of the constraints and the dual variables p_j with $j \in J$ for the second part.

$$
\begin{aligned}
(P) \quad \underset{\text{s.t.}}{\text{Max}} \sum_{i \in I} -s_i \pi_i + \sum_{j \in J} d_j p_j & \\
p_j - \pi_i \leq c_{ij}, & \quad \forall i \in I,\, j \in J \\
\pi_i, p_j \geq 0, & \quad \forall i \in I,\, j \in J
\end{aligned}
$$

Note: Each x_{ij} occurs exactly once with coefficient 1 in both parts of the constraints, resulting in the dual constraints.

c) An external transport company offers to take over the transport and wants to maximize its profit. The dual variable π_i for each collection point i can be interpreted as the price the transportation company has to pay in order to buy food at the these locations. The variables p_j correspond to the selling price at foodbanks j. The company wants to ensure that the profit $p_j - \pi_i$ does not exceed the transportation costs c_{ij} between these locations, as otherwise you would not hire the company and could take care of the transportation yourself.